

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1613

Mesa 138 kV
December 03, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1613. The Interconnection Customer proposes a 88.1 MW battery energy storage facility interconnecting at the Mesa 138 kV substation in the Virginia area. The request was received on November 22, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 589 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,017,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dps Ocean Spray-Windsor Hills 138 KV	Short Pump-S Meadow 2 138 KV L	130.7	230	36.4
Short Pump-Mesa 138 KV	Souhwick-South Meadow 2 138 KV	128.6	288	67.5
Windsor Hills-Southwick 138 KV	Short Pump-South Meadow 2 138 KV	126.4	260	36.4
Dps Ocean Spray-Pipers Gap 138 KV	Leavitt-Birchwood 345 KV	124.4	229	37.2
S Meadow 2-Southwick 138 KV Line	Short Pump-S Meadow 2 138 KV Line	123.2	288	36.4
Short Pump-South Meadow 2 Substation 138 KV L	Southwick-South Meadow 2 138 KV	121.7	359	67.5
Mesa-Pipers Gap Substation 138 KV	Leavitt-Birchwood 345 KV	121.0	257	37.2
Bear-Fluvanna P.S 345 KV	Leavitt-Birchwood 345 KV	104.6	1,663	13.7
Nestuca-Polk City 345 KV	Nauvoo-Birchwood Substation 345 KV	100.4	1,458	5.4
Birchwood-Nauvoo Substation 345 KV	Fluvanna P.S-Louisa Generation Facility 345 KV	100.0	2,444	18.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dps Ocean Spray-Windsor Hills 138 KV	Rebuild line with higher-capacity conductor	\$7,008,000	42% (\$2,909,000)
Short Pump-Mesa 138 KV	Rebuild line with higher-capacity conductor	\$2,905,000	54% (\$1,571,000)
Windsor Hills-Southwick 138 KV	Rebuild line with higher-capacity conductor	\$7,245,000	42% (\$3,007,000)
Dps Ocean Spray-Pipers Gap 138 KV	Rebuild line with higher-capacity conductor	\$2,785,000	46% (\$1,267,000)
S Meadow 2-Southwick 138 KV Line	Rebuild line with higher-capacity conductor	\$1,470,000	42% (\$610,000)
Short Pump-South Meadow 2 Substation 138 KV L	Rebuild line with higher-capacity conductor	\$3,529,000	54% (\$1,908,000)
Pipers Gap-Mesa 138 KV line	Rebuild line with higher-capacity conductor	\$2,924,000	46% (\$1,330,000)
Bear-Fluvanna P.S 345 KV	Reconductor line	\$29,044,000	1% (\$179,000)
Nestuca-Polk City 345 KV	Reconductor line	\$9,880,000	1% (\$65,000)
Birchwood-Nauvoo Substation 345 KV	Reconductor line	\$20,976,000	1% (\$171,000)
	Total allocated to project		\$13,017,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.